

Kheem Parkash Dharmani

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SUMMARY

Lead AI Engineer building production LLM and retrieval-augmented generation (RAG) systems for clinicians, with an MS in Data Science and research in clinical NLP, radiology report generation and LLM evaluation. Seeking a fully funded PhD on rigorous evaluation of generative AI in healthcare: faithfulness to evidence, uncertainty and calibration, and robustness across clinical settings.

WORK EXPERIENCE

Lead AI Engineer at Clinical Pearl, Princeton, NJ (remote) Apr 2025 – Present

- Lead a team of three engineers building a production RAG platform for clinical decision support, serving 1,000+ clinicians.
- Reduced query latency by 40%, improved answer relevance by 20%+ and cut deployment failures by 35%.
- Designed LLM evaluation and observability tooling; developed the CAMEO-AI and PEARL-G evaluation frameworks (presented at EBM Live, Oxford, June 2026).
- Hire and interview engineers; work with HIPAA-regulated protected health information (PHI).

ML / AI Engineer at Clinical Pearl, Princeton, NJ (remote) Apr 2024 – Mar 2025

- Fine-tuned large language models (e.g., OpenAI GPT, LLaMA, Mistral) for randomized clinical trial applications.
- Designed and deployed RAG systems for disease and pharmaceutical information retrieval, using PostgreSQL/pgvector.
- Built multimodal RAG systems integrating structured and unstructured data for medical decision-making.

Machine Learning Engineer at Cplus Soft, Islamabad Jan 2023 – Mar 2024

- Part-time Jan–Nov 2023; full-time Dec 2023 – Mar 2024.
- Developed generative AI solutions for question answering and text summarization, improving task performance by 10–30% across use cases.
- Integrated LangChain into AI applications, improving the modularity and scalability of workflows by 25%.

Machine Learning Trainer at AI Lounge, Gilgit-Baltistan Jul 2023 – Nov 2023

- Designed and delivered a Data Science and AI curriculum covering machine learning, deep learning and data visualization.

NLP Teaching Assistant at FAST-NUCES, Islamabad Jan 2023 – Jun 2023

- Simplified complex NLP concepts for students (95% course satisfaction) and mentored students on NLP tasks.

EDUCATION

MS Data Science, FAST-NUCES, Islamabad 2022 – 2024 (CGPA 3.20/4.00)

Thesis: *A Hybrid Approach for Radiology Report Generation using Vision Transformers & Language Models* (MIMIC-CXR); improved BLEU by 45% and ROUGE-L by 55%.

BE Computer Systems Engineering, Mehran UET, Jamshoro 2016 – 2020 (CGPA 3.72/4.00)

RESEARCH AND PUBLICATIONS

- K. R. Narejo, H. Zan, **K. P. Dharmani**, O. Mamyrbayev, A. Akhmediyarova, Z. Alibiyeva, J. Alimkulova. *Optimizing Sentiment Integration in Image Captioning Using Transformer-Based Fusion*

Strategies. Computers, Materials & Continua 84(2):3407–3429, 2025.

- CAMEO-AI and PEARL-G: evaluation frameworks for clinical AI, presented at EBM Live, Oxford, June 2026.
- K. R. Narejo, H. Zan, **K. P. Dharmani**, L. Zhou, T. J. Alahmadi, N. Sehito, Y. Y. Ghadi, M. Assam. *EEBERT: An Emoji-Enhanced BERT Fine-Tuning on Amazon Product Reviews for Text Sentiment Classification*. *IEEE Access*, 2024.
- K. R. Narejo, H. Zan, D. Oralbekova, **K. P. Dharmani**, O. Mamyrbayev, K. Mukhsina. *Enhancing Emoji-Based Sentiment Classification in Urdu Tweets: Fusion Strategies with Multilingual BERT and Emoji Embeddings*. *IEEE Access*, 2024.

PROJECTS

Medical Knowledge Retrieval and Question-Answering System – FastAPI application using RAG over structured medical data (OpenAI, LangChain); improved retrieval efficiency by 40%.

Nabz – AI health companion (Alibaba Cloud AI Hackathon 2026): Urdu voice triage, lab-report and prescription reading with Qwen-VL, family Medical Vault; React, FastAPI, Qwen.

Control Automation Agent – LangGraph agent that tests IT access controls over full evidence populations with an evaluation harness; FastAPI, Streamlit.

ETL Pipeline for Churn Prediction on MS Azure – end-to-end pipeline from OLTP streams to an Azure SQL database with Azure Data Factory and Streamlit ([GitHub](#)).

SKILLS

Programming	Python, R, SQL
ML / DL	PyTorch, TensorFlow, Keras, scikit-learn; CNNs, RNNs, LSTMs, autoencoders
NLP / LLMs	LLMs, transformers, RAG, LangChain, LLM evaluation and observability, BERT, spaCy, NLTK
Data & Backend	PostgreSQL / pgvector, FastAPI, Streamlit
Cloud	MS Azure (Data Factory, Cognitive Services); AWS Solutions Architect course (SOOFTO, Bahria University)
Tools	Git, GitHub, Tableau, Matplotlib, Seaborn

CERTIFICATIONS AND AWARDS

- Machine Learning Specialization (Dec 2023) and Advanced Learning Algorithms (Oct 2023) – DeepLearning.AI; Supervised Machine Learning: Regression and Classification – Stanford Online (Mar 2023); Python for Data Science, AI & Development – IBM (Mar 2023).
- IEEE eLearning Course Trial Award, IEEE Xplore Challenge for Researchers in Pakistan (Sep 2023), endorsed by FAST-NUCES Islamabad.